

Shower project

Team OH-J

Element I: Testing, Data Collection, and Analysis

In the boys dorms the dorm residents need a place to put their toiletries because there is not a built-in shelf in the communal showers and setting things on the floor is unsanitary. Solving the problem can help the dorm students place their toiletries somewhere without having to set it down on the ground. The problem is that there is nowhere to hang your shampoo and that leads to people putting it on the ground.

The testing procedures we are putting our product through will give us information on whether our product passes our high priority requirement. Our high priority requirements are as follows: stable (does not fall when bumped into) , waterproof (is not damaged by water and water does not get caught inside the design), durability (In the case the caddy falls from our elevated height of 4 feet it will not break), capacity (how many bottles can fit inside the caddy), portability (is able to be carried easily with one hand).

For the first test we will be testing stability by putting it up against the wall and walking into it to make sure it doesn't fall down. We tested it four times, two with bottles and two without.

Tests	Test 1: no bottles	Test 2: no bottles	Test 3: with bottles	Test 4: with bottles
Success or Failure	Success	Success	Success	Success

For our second test we will test whether our product is waterproof or not by seeing if our design when put under a water source will drain efficiently or fill with water. The way our team will test the requirement is by placing it under a large sink spout and running the water into the caddy and seeing if it fills with water whatsoever we will test this three times.

Test	Test 1	Test 2	Test 3
Success or Failure	Success	Success	Success

The next test that will be performed is a durability test. The team will test the durability by dropping the caddy from the 4 varying heights starting with the height the caddy will be placed at in the shower (four feet elevation off the floor) and increase by one foot after each test.

Height of elevation of test drop	4 feet	5 feet	6 feet	7 feet
Success or Failure	Success	Success	Success	Success

Capacity test, the parameters of the test are to put as many varying size bottles as can fit in the caddy. The test results are 40 travel soups/small, 6 medium bottles, and 5 large bottles. The next test is a portability test to see if the caddy is able to be moved easily with one hand. To test this one member of the team will carry the caddy in one hand back and forth across the classroom.